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Title- The Illusion of Improvement: Insights from a Digital Marketing A/B Test

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Abstract

A/B testing is a widely used framework in digital marketing for evaluating alternative campaign strategies through controlled experimentation. While improvements in engagement metrics such as click through rate CTR are often interpreted as indicators of success, they may not necessarily translate into meaningful business outcomes. This study analyses a digital advertising experiment conducted during August 2019, comparing a Control and a Test campaign across multiple stages of the customer conversion funnel. Using statistically appropriate methods including two sample Z tests for proportion-based metrics and a Welch two sample t test for cost efficiency, we examine differences in click through rate, conversion rate, and cost per purchase. The results show that although the Test campaign achieves a significantly higher CTR, this increase in engagement does not lead to improved conversion efficiency or superior cost effectiveness. The findings highlight the importance of funnel level evaluation and caution against relying on isolated metrics when making data driven marketing decisions.

Keywords- A/B Testing, Customer Funnel, Click-through Rate, Conversion Rate, Cost Per Purchase

The Illusion of Improvement: Insights from a Digital Marketing A/B Test

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Introduction

Digital advertising campaigns aim to influence consumer behaviour across multiple stages, ranging from initial exposure to final purchase. A/B testing provides a systematic approach for evaluating whether modifications in campaign designs lead to improved performance relative to an existing baseline.

In practice, campaign success is frequently judged using engagement-based metrics such as click through rate (CTR). However, higher engagement does not necessarily imply increased revenue or profitability. Decisions based solely on early-stage metrics may therefore lead to misleading conclusions. This study adopts a funnel-based perspective by evaluating campaign performance sequentially across engagement, conversion, and cost-efficiency metrics, thereby providing a more comprehensive assessment of campaign effectiveness.

Data Description and Experimental Setup

The dataset consists of daily observations from two digital advertising campaigns conducted concurrently during August 2019:

- **Control Campaign**: baseline advertisement
- **Test Campaign**: modified advertisement with revised creative elements

Both campaigns targeted similar audience segments to ensure comparability. The dataset includes the following variables:

- Impressions
- Website Clicks
- Purchases
- Spend (USD)
- Additional funnel indicators such as searches, view content, and add-to-cart events

The dataset contains **59 daily observations for each campaign**. All variables are discrete counts, except for spend, which is continuous. Given the limited temporal scope and single-platform setting, the findings are illustrative rather than generalizable.

Statistical Framework

At each funnel stage, outcomes are modelled as Bernoulli trials:

- An impression results in a click or not
- A click results in a purchase or not

Aggregated outcomes therefore follow Binomial distributions. With large sample sizes, sample proportions are approximately normally distributed, motivating the use of two-sample Z-tests for proportion-based metrics (Spiegel et al., 2013).

Cost-based metrics such as cost per purchase are continuous and not proportion-based; hence, alternative statistical tests are required. Here we have used Welch two sample t-test.

Stage I: Click-Through Rate (CTR)

Definition

$$\text{CTR} = \frac{\text{Number of Website Clicks}}{\text{Number of Impressions}}$$

CTR measures the ability of an advertisement to attract user attention.

Hypotheses

Let p_C and p_T denote the true CTRs of the Control and Test campaigns.

$$H_0: p_T = p_C$$
$$H_1: p_T > p_C$$

Test Statistic

The Test statistic (Z) is given by:

$$Z = \frac{\hat{p}_T - \hat{p}_C}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_T} + \frac{1}{n_C}\right)}}$$

Computations

From the dataset:

Campaign	Total Clicks	Total Impressions
Control	154303	3177233
Test	180970	2237544

We compute the sample proportions (CTR). We observe $\hat{p}_C = 0.0486$ and $\hat{p}_T = 0.0809$

Pooled proportion is given as:

$$\hat{p} = \frac{x_C + x_T}{n_C + n_T} = \frac{154303 + 180970}{3177233 + 2237544} = 0.0619$$

where x denotes the Total Clicks of the Control or Test campaign and n denotes the total impressions of the Control or Test Campaign.

Substituting the values we get

$$Z \approx 153.63$$

At significance level $\alpha = 0.05$;

$$z_{0.05} = 1.645$$

Since $Z \approx 153.63 > 1.645 = z_{0.05}$

Hence, we **reject** H_0.

After conducting the two-sample Z-test for difference in proportions, the Test campaign exhibited a substantially higher sample CTR than the Control campaign, yielding a very large positive Z-statistic and leading to rejection of the null hypothesis at the 5% significance level.

The Test campaign is statistically more effective at generating clicks than the Control campaign. This result indicates stronger user engagement at the advertisement level. However, CTR captures only the initial stage of interaction and provides no direct information about downstream purchasing behaviour. Therefore, conclusions regarding revenue or profitability cannot be drawn at this stage.

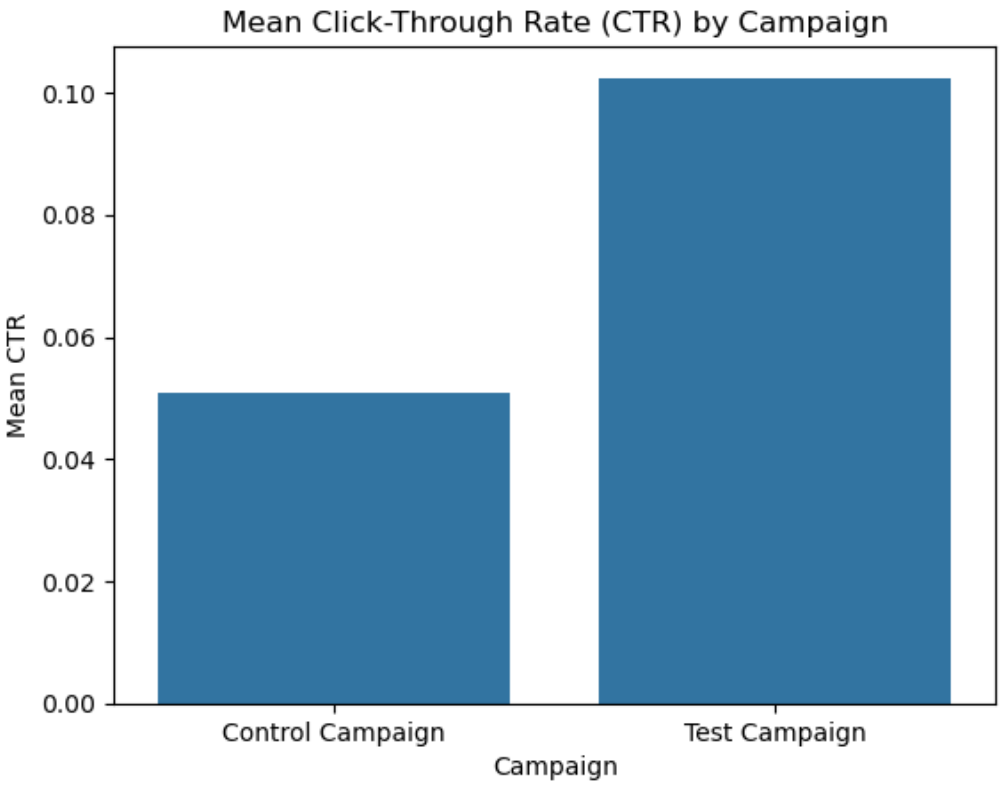

Figure 1: The figure suggests that the Test campaign achieved a significantly higher CTR compared to the Control Campaign that implies that the modified advertisement was more effective at attracting user attention and generating customer engagement.

From Engagement to Economic Action

To evaluate this engagement translates into a meaningful economic, we consider Conversion Rate as the next performance metric. This captures how appealing an offer is to persuade a customer to finally purchase that product.

Stage II: Conversion Rate (CR)
Definition

$$\text{CR} = \frac{\text{Number of Purchases}}{\text{Number of Website Clicks}}$$

Conversion rate measures the effectiveness of converting interested users into paying customers.

Hypotheses
Let p_C and p_T denote the true conversion rates of the Control and Test campaigns.

$$H_0: p_T = p_C$$
$$H_1: p_T > p_C$$

From the dataset:

Campaign	Total Purchases	Total Clicks
Control	15161	154303
Test	15637	180970

The sample proportions (CR) are $\hat{p}_C = 0.0982$ and $\hat{p}_T = 0.0864$.
Pooled proportion is

$$\hat{p} = \frac{y_C + y_T}{n_C + n_T} = 0.0919$$

Where y represents the Total Purchases of the Control or Test campaign.

Here, test statistic Z = -11.837.

At significance level $\alpha = 0.05$; $z_{0.05} = 1.645$.

Here, $Z \approx -11.837 < \ 1.645 = z_{0.05}$; thus we **accept** H_0.

Similarly, like CTR, here also a two-sample Z-test for proportions was performed on the conversion rates. The resulting test statistic was negative and statistically significant in magnitude, indicating that the Test campaign does not outperform the Control campaign in terms of conversion efficiency.

Despite attracting more users, the Test campaign does not achieve a higher conversion rate, suggesting that the additional traffic may have lower purchase intent.

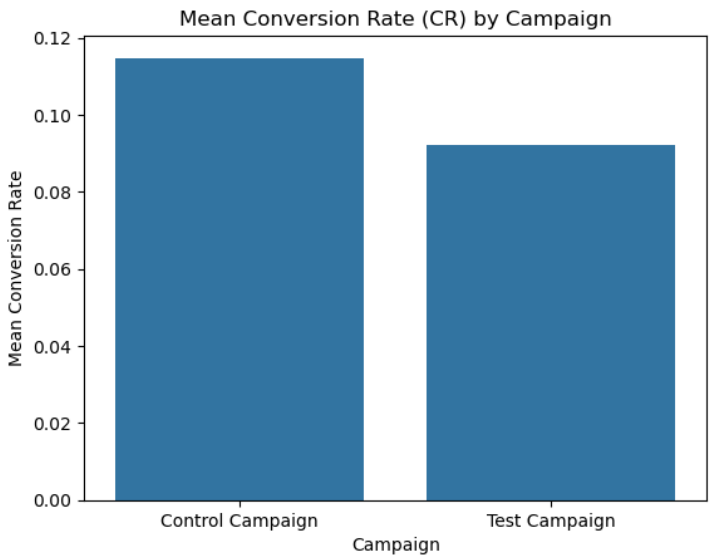

Figure 2: The figure compares the effectiveness of the Control and Test Campaign in converting website clicks into actual purchases. The test campaign generated more engagement but lower conversion rate in comparison to the Control Campaign

The Price behind Performance

To gain a knowledge about the true profitability of the campaign we must move beyond conversion outcomes and focus on the financial efficiency. This is captured by the Cost Per Purchase (CPP) which provides direct insight into the economic effectiveness of the campaign. It measures how much it costs to get a single sale.

Definition

$$\text{CPP} = \frac{\text{Total Spend}}{\text{Number of Purchases}}$$

Cost per purchase measures the financial efficiency of a campaign and is directly related to profitability(Investopedia, 2024).

CPP is a continuous metric derived from two random variables and does not follow a Binomial structure. Consequently, Z-tests for proportions are inappropriate.
To compare CPP between campaigns, a **Welch two-sample t-test** is employed due to:

- Continuous nature of CPP
- Potential inequality of variances
- Robustness to non-normality under moderate sample sizes

The hypotheses are formulated as:

$$H_0: \mu_T = \mu_C$$
$$H_1: \mu_T < \mu_C$$

where $\mu_C$and $\mu_T$denote the mean CPP of the Control and Test campaigns.
The alternative hypothesis reflects the business objective that **lower CPP is preferable**.

Test Statistic (Welch t-statistic)

$$t = \frac{\bar{Y} - \bar{X}}{\sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}}}$$

Where
$\bar{X}$, s_C^2be the sample mean and variance of CPP for Control

$\bar{Y}$, s_T^2be the sample mean and variance of CPP for Test

At significance level $\alpha = 0.05$:
Reject H_0 iff

$$t < t_{\alpha,\nu}$$

The degrees of freedom are approximated using the Welch–Satterthwaite equation:

$$\nu = \frac{\left(\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}\right)^2}{\frac{(s_T^2/n_T)^2}{n_T - 1} + \frac{(s_C^2/n_C)^2}{n_C - 1}}$$

From the dataset:

Campaign	**Mean CPP ($\bar{x}$)**	**Standard Deviation(s)**	**Sample Size(n)**
Control	5.05	2.12	29
Test	5.90	2.80	30

Here t ≈ 1.32 ; $\nu = 53.8 \approx 54$

At significance level $\alpha = 0.05$; $t_{\alpha,\nu} \approx -1.67$

Since, t > $t_{\alpha,\nu}$ so we **accept** H_0 .

At 5% significance level, the given data provides extremely strong evidence to conclude the CPP of the Control campaign is same as that of the Test Campaign. This indicates that the test campaign does not achieve a statistically significant improvement in cost efficiency in comparison to the Control campaign.

Since the Test campaign does not improve conversion efficiency, higher engagement is unlikely to improve cost performance.

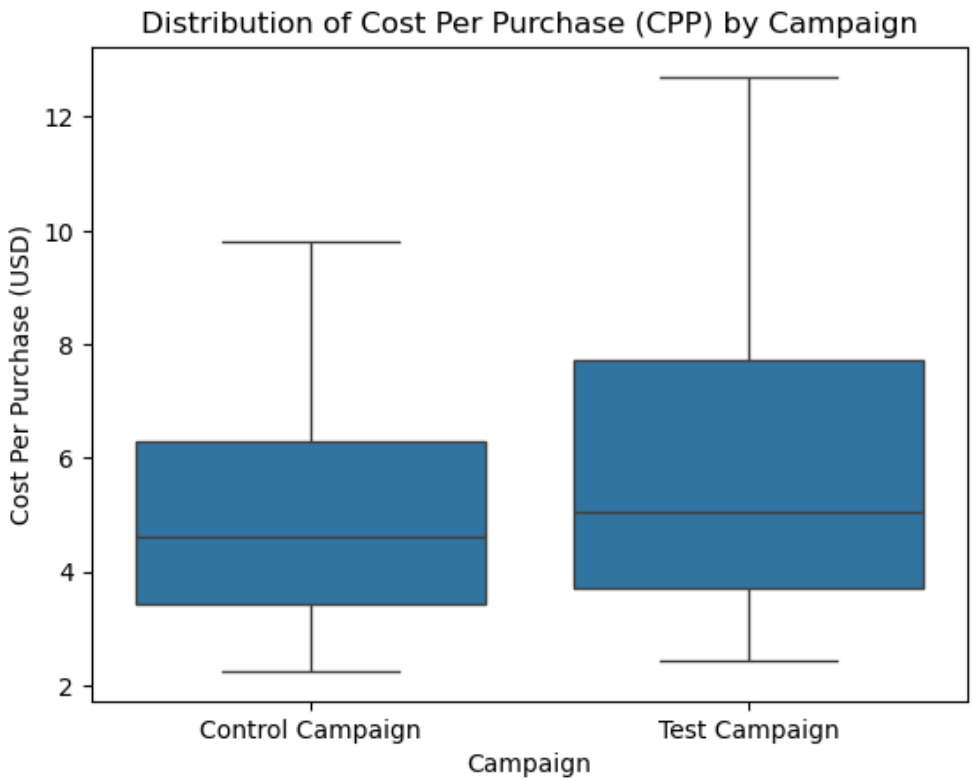

Figure 3: The boxplot above shows that the Test campaign has a higher median CPP in comparison to the Control Campaign indicating higher acquisition cost. The Test campaign also has a higher variance indicated by its larger box size and longer whiskers. The visual spread also suggests that not only higher on average but more prone to inconsistencies.

Strategic Divergence: A Managerial Review

The empirical findings indicate that the modified advertisement is effective at attracting customer attention but fails to increase purchase intent or reduce the customer acquisition cost. Hence the Test Campaign does generate additional activity at the top of the funnel but does not deliver corresponding conversion value or profitability.

From a managerial perspective this divergence highlights the risk of equating increased visibility with the actual business success. Improvements in engagement metrics might imply that the campaign is performing better, which can lead to hasty scaling or shifting budgets.

However, reallocating resources based solely on these metrics may lead to suboptimal decision-making if downstream outcomes are not simultaneously evaluated.

Limitations

This study is subject to several limitations:

- The analysis is based on **59 daily observations from a single month (August 2019)**
- Results are **platform-specific and time-bound**
- Seasonal effects and cross-industry variability are not captured

Accordingly, the findings should be viewed as exploratory, illustrating methodological principles rather than universal marketing rules.

Conclusion

This study demonstrates that effective A/B testing requires metric-specific statistical testing and funnel-level interpretation. While the Test campaign significantly improves user engagement, it fails to deliver superior conversion or cost efficiency. These findings reinforce the need for holistic evaluation frameworks in digital marketing and caution against over-reliance on isolated engagement metrics when making strategic decisions.

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